

1. Administrative & Study Identification

Full Study Title: A Phase I/II Study on Safety AND Immunogenicity of AZD4117 and AZD5315 Vaccines **Short Title/Acronym:** PANDA Trial **Protocol Number:** D8500C00001 **NCT Number:** NCT07128615 **Sponsor Name:** AstraZeneca **Investigational Product:** AZD4117 and AZD5315 Vaccines **Phase of Development:** Phase I/II **Medical Monitor:** [AstraZeneca Medical Monitor - Contact Information]

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety and immunogenicity of two investigational vaccines, AZD4117 and AZD5315, to protect against certain strains of avian Influenza A (H5N1 and H7N9 subtypes). **Secondary Objectives:** To assess whether participants achieve protective antibody levels (hemagglutination inhibition [HAI] $\geq 1:40$ titer) and measure seroconversion rates; to evaluate optimal dosage levels across different adult age groups.

2.2 Background & Rationale This study represents a significant advancement in flu prevention, specifically designed to protect against Influenza A, a serious respiratory illness that causes seasonal outbreaks and can lead to severe complications. The research targets evolving avian influenza strains (H5N1 and H7N9) to ensure optimal protection, especially in populations like older adults who may experience diminished immune responses.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** Randomized

3.2 Planned Enrollment & Duration Target \$N\$: Approximately 400 healthy adult volunteers (ages 18-64 and 65+) **Number of Sites:** Multiple sites in the USA **Estimated Study Start:** 04-Sep-2025 **Estimated Primary Completion:** 28-Jan-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adults, ≥ 18 years of age at the time of signing the informed consent. 2. Participants who are medically stable such that, according to the judgement of the Investigator, hospitalization within the study period is not

anticipated. 3. Written informed consent and any locally required authorization obtained prior to performing any protocol-related procedures.

4.2 Exclusion Criteria 1. Receipt of immune-modifying drugs, immunosuppressive therapy, or long-term systemic corticosteroid therapy within 6 months prior to enrollment. 2. Acute (time-limited) or febrile (temperature $\geq 38.0^{\circ}\text{C}$) illness/infection within 3 days of intended IMP administration. 3. Individuals who have had a previous confirmed or suspected illness from influenza caused by an H5N1 or H7N9 virus.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Different dosage levels of AZD4117 or AZD5315, or matched placebo injection. **Route of Administration:** Intramuscular (IM) injection. **Duration of Treatment:** Specific vaccination schedule with follow-up spanning up to the estimated study completion in December 2026.

5.2 Concomitant & Prohibited Therapy Permitted: Topical/inhaled steroids or short-term oral steroids. **Prohibited:** Receipt of immunoglobulin or blood products within 6 months prior; immune-modifying drugs or immunosuppressive therapy (e.g., anti-cancer chemotherapy/radiation); long-term systemic corticosteroids.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -14 to 0	Informed Consent, Medical History, Baseline Health Screening
Baseline	Day 0	Randomization, Vaccination (AZD4117, AZD5315, or Placebo), Immediate Reaction Monitoring (30 mins)
Interim	Post-Vaccination	Safety follow-ups, Immunogenicity blood draws for antibody titers
End of Study	Follow-up End	Final Vital Signs, Exit Interview, Final Immunogenicity/Safety Review

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Assessment of vaccine safety profiles and overall immunogenicity (protective antibody production against Influenza A H5N1 and H7N9). **Measurement Method:** Clinical safety monitoring and standard laboratory measures of vaccine effectiveness.

7.2 Secondary & Exploratory Endpoints * Hemagglutination inhibition (HAI) antibody titers ($\geq 1:40$ titer) indicating protective levels. * Seroconversion rates across different age cohorts (18-64 vs. ≥ 65 years).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Close monitoring for immediate reactions within 30 minutes post-vaccination, followed by patient-reported safety diaries and scheduled follow-ups. **Serious Adverse Events (SAEs):** Standard reporting timeline, typically within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Independent safety monitoring mechanisms managed per AstraZeneca and ICH GCP guidelines.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT), Per-Protocol (PP), and Safety Set populations. **Sample Size Justification:** Target enrollment of roughly 400 individuals is powered to adequately evaluate safety differences and immunological seroconversion across two specific age strata. **Handling of Missing Data:** Standard handling typically utilizing Multiple Imputation (MI) or Last Observation Carried Forward (LOCF) for clinical endpoints.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine clinical site monitoring and source data verification. **Audit Trail:** Compliant eCRF data entry, locking procedures, and tracking of protocol deviations.

11. Study Identifiers & Governance

Primary ID: D8500C00001 **Registry Number:** NCT07128615 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** PANDA **Official Regulatory Title:** A Phase I/II Study on Safety AND Immunogenicity of AZD4117 and AZD5315 Vaccines (PANDA)

12. Clinical Framework (Study Specific)

Primary Condition: Influenza A (H5N1 and H7N9 subtypes) prevention **Diagnosis Threshold:** Healthy adults requiring prophylactic vaccination **Medical Methodology:** Investigational vaccine administration **Optimization Target:** Achievement of protective antibody levels (HAI $\geq$ 1:40 titer) and seroconversion

Management Pathway: Double-blinded, randomized vaccination protocol
Education Protocol (HCP): Clinical trial protocol adherence and post-vaccine safety reaction monitoring
Education Protocol (Patient): Informed consent outlining trial design, vaccine administration, and symptom reporting
Quality Audit Mechanism: Independent clinical monitoring, blinded safety review, and centralized laboratory assessments

Total Duration: Primary completion in roughly 20 weeks (Sep 2025 to Jan 2026), full follow-up until Dec 2026 **Onsite Visit Cadence:** Pre-defined screening, vaccination, and immunogenicity blood-draw visits **Remote Monitoring Frequency:** Periodic safety check-ins and symptom reporting intervals **Checklist Review Interval:** Regular review of patient symptom diaries post-vaccination

Role	Name	Signature	Date	---	---	---	---
Principal Investigator	[Insert Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Insert Name]	_____	[DD-MMM-YYYY]				
Lead Statistician	[Insert Name]	_____	[DD-MMM-YYYY]				